

# Internal Dependencies

## References

- [Analyze java package metrics in a graph database](#)
- [Calculate metrics](#)
- [Neo4j Python Driver](#)

## Artifacts

List the artifacts this notebook is based on. Different sorting variations help finding artifacts by their features and support larger code bases where the list of all artifacts gets too long.

Only the top 30 entries are shown. The whole table can be found in the following CSV report:

List\_all\_Java\_artifacts

Table 1a - Top 30 artifacts with the highest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 1  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 3  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 5  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 8  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 10 | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |

Table 1b - Top 30 artifacts with the highest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 1  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 2  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 3  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 7  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 8  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 9  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 10 | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |

Table 1c - Top 30 artifacts with the highest number of incoming dependencies

The following table lists the top 30 artifacts that are used the most by other artifacts (highest count of incoming dependencies, highest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 1  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 2  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 3  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 4  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 5  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 8  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 9  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 10 | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |

## Table 1d - Top 30 artifacts with the highest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the highest number of other artifacts (highest count of outgoing dependencies, highest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 1  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 2  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 3  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 4  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 6  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 7  | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |
| 8  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 10 | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |

## Table 1e - Top 30 artifacts with the lowest package count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 2  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 3  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 4  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 6  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 7  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 8  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 9  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 10 | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |

## Table 1f - Top 30 artifacts with the lowest type count

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |
| 1  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 2  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 4  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 5  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 6  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 7  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 8  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 9  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 10 | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |

## Table 1g - Top 30 artifacts with the lowest number of incoming dependencies

The following table lists the top 30 artifacts that are used the least by other artifacts (lowest count of incoming dependencies, lowest in-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 1  | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |
| 2  | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |
| 3  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 4  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 5  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 6  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 7  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 8  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 9  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 10 | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |

## Table 1h - Top 30 artifacts with the lowest number of outgoing dependencies

The following table lists the top 30 artifacts that are depending on the lowest number of other artifacts (lowest count of outgoing dependencies, lowest out-degree).

|    | artifactName                             | packages | types | incomingDependencies | outgoingDependencies |
|----|------------------------------------------|----------|-------|----------------------|----------------------|
| 0  | axon-common-5.0.1.jar                    | 13       | 150   | 10                   | 0                    |
| 1  | axon-conversion-5.0.1.jar                | 4        | 30    | 4                    | 1                    |
| 2  | axon-update-5.0.1.jar                    | 5        | 23    | 1                    | 1                    |
| 3  | axon-messaging-5.0.1.jar                 | 57       | 570   | 7                    | 2                    |
| 4  | axon-metrics-micrometer-5.0.1.jar        | 2        | 13    | 0                    | 2                    |
| 5  | axon-tracing-opentelemetry-5.0.1.jar     | 1        | 5     | 0                    | 2                    |
| 6  | axon-modelling-5.0.1.jar                 | 7        | 93    | 2                    | 3                    |
| 7  | axon-test-5.0.1.jar                      | 5        | 73    | 1                    | 3                    |
| 8  | axon-eventsourcing-5.0.1.jar             | 7        | 100   | 3                    | 4                    |
| 9  | axon-server-connector-5.0.1.jar          | 5        | 72    | 1                    | 4                    |
| 10 | axon-spring-boot-autoconfigure-5.0.1.jar | 7        | 70    | 0                    | 7                    |

## Cyclic Dependencies

Cyclic dependencies occur when one package uses a class of another package and vice versa. These dependencies can lead to problems when one of these packages needs to be changed.

## Table 2a - Cyclic Dependencies Overview

Show the top 40 cyclic dependencies sorted by the most promising to resolve first. This is done by calculating the number of forward dependencies (first cycle participant to second cycle participant) in relation to backward dependencies (second cycle participant back to first cycle participant). The higher this rate (approaching 1), the easier it should be to resolve the cycle by focussing on the few backward dependencies.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

[Cyclic\\_Dependencies](#)

### Columns:

- *artifactName* identifies the artifact of the first participant of the cycle
- *packageName* identifies the package of the first participant of the cycle
- *dependentArtifactName* identifies the artifact of the second participant of the cycle
- *dependentPackageName* identifies the package of the second participant of the cycle
- *forwardToBackwardBalance* is between 0 and 1. High for many forward and few backward dependencies.
- *numberForward* contains the number of dependencies from the first participant of the cycle to the second one
- *numberBackward* contains the number of dependencies from the second participant of the cycle back to the first one
- *someForwardDependencies* lists some forward dependencies in the text format "type1 -> type2"
- *backwardDependencies* lists the backward dependencies in the format "type1 <- type2" that are recommended to get resolved

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | forwardToBackwardBalance | numberForward | numberBack |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|--------------------------|---------------|------------|
| 0  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | 0.959184                 | 48            |            |
| 1  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | 0.942857                 | 34            |            |
| 2  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | 0.933333                 | 29            |            |
| 3  | axon-messaging-5.0.1        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | 0.894737                 | 18            |            |
| 4  | axon-messaging-5.0.1        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | 0.875000                 | 15            |            |
| 5  | axon-modelling-5.0.1        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.1        | org.axonframework.modelling                             | 0.846154                 | 12            |            |
| 6  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 7  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | 0.666667                 | 15            |            |
| 8  | axon-common-5.0.1           | org.axonframework.common.configuration                                | axon-common-5.0.1           | org.axonframework.common.infra                          | 0.529412                 | 13            |            |
| 9  | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | 0.500000                 | 15            |            |
| 10 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork             | 0.500000                 | 6             |            |
| 11 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | 0.428571                 | 5             |            |
| 12 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.command          | 0.333333                 | 4             |            |
| 13 | axon-messaging-5.0.1        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | 0.333333                 | 2             |            |
| 14 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling               | 0.333333                 | 8             |            |
| 15 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | 0.200000                 | 6             |            |
| 16 | axon-modelling-5.0.1        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.1        | org.axonframework.modelling.annotation                  | 0.200000                 | 3             |            |
| 17 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork                           | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | 0.157895                 | 11            |            |
| 18 | axon-modelling-5.0.1        | org.axonframework.modelling.entity.child                              | axon-modelling-5.0.1        | org.axonframework.modelling.entity                      | 0.142857                 | 8             |            |
| 19 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.util                           | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | 0.000000                 | 2             |            |
| 20 | axon-eventsourcing-5.0.1    | org.axonframework.eventsourcing.annotation.reflection                 | axon-eventsourcing-5.0.1    | org.axonframework.eventsourcing.annotation              | 0.000000                 | 1             |            |

Table 2b - Cyclic Dependencies Break Down

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown

Columns in addition to Table 2a:

- *dependency* shows the cycle dependency in the text format "type1 -> type2" (forward) or "type2<-type1" (backward)

[illegible]

Table 2c - Cyclic Dependencies Break Down - Backward Dependencies Only

Lists packages with cyclic dependencies with every dependency in a separate row sorted by the most promising dependency first. This table only contains the backward dependencies from the second participant of the cycle back to the first one that are the most promising to resolve.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Cyclic\_Dependencies\_Breakdown\_BackwardOnly

|    | artifactName                | packageName                                                           | dependentArtifactName       | dependentPackageName                                    | dependency forName                                                           |
|----|-----------------------------|-----------------------------------------------------------------------|-----------------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| 0  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation                           | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | HandlerAttributes<-SimpleHandlerAttributes                                   |
| 1  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling                             | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | EventMessage<-SubscribableEventSource                                        |
| 2  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.annotation                  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | EventHandler<-HandlerTypeResolver                                            |
| 3  | axon-messaging-5.0.1        | org.axonframework.messaging.commandhandling.annotation                | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | CommandHandler<-HandlerTypeResolver                                          |
| 4  | axon-messaging-5.0.1        | org.axonframework.messaging.queryhandling.annotation                  | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | QueryHandler<-HandlerTypeResolver                                            |
| 5  | axon-modelling-5.0.1        | org.axonframework.modelling.annotation                                | axon-modelling-5.0.1        | org.axonframework.modelling                             | TargetEntityIdMemberMismatchException<-PropertyBasedEntityIdResolver         |
| 6  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorsConfigurer<-EventProcessingConfigurer          |
| 7  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorModule<-EventProcessorModule                    |
| 8  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.streaming.pooled | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | PooledStreamingEventProcessorConfiguration<-EventProcessorModule             |
| 9  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorConfiguration<-EventProcessorModule                 |
| 10 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorModule<-EventProcessorModule                        |
| 11 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.processing.subscribing      | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | SubscribingEventProcessorsConfigurer<-EventProcessingConfigurer              |
| 12 | axon-common-5.0.1           | org.axonframework.common.configuration                                | axon-common-5.0.1           | org.axonframework.common.infra                          | Component<-FilesystemStyleComponentDescriptor                                |
| 13 | axon-common-5.0.1           | org.axonframework.common.configuration                                | axon-common-5.0.1           | org.axonframework.common.infra                          | Component\$Identifier<-FilesystemStyleComponentDescriptor                    |
| 14 | axon-common-5.0.1           | org.axonframework.common.configuration                                | axon-common-5.0.1           | org.axonframework.common.infra                          | Component\$Identifier<-JacksonComponentDescriptor                            |
| 15 | axon-common-5.0.1           | org.axonframework.common.configuration                                | axon-common-5.0.1           | org.axonframework.common.infra                          | Component<-JacksonComponentDescriptor                                        |
| 16 | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-ChainedMessageHandlerInterceptorMember |
| 17 | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-AnnotatedHandlerInspector                                |
| 18 | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | MessageHandlerInterceptorMemberChain<-AnnotatedHandlerInspector              |
| 19 | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | MessageInterceptingMember<-AnnotatedHandlerInspector                         |
| 20 | axon-messaging-5.0.1        | org.axonframework.messaging.core.interception.annotation              | axon-messaging-5.0.1        | org.axonframework.messaging.core.annotation             | NoMoreInterceptors<-ChainedMessageHandlerInterceptorMember                   |
| 21 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork             | TransactionManager<-TransactionalUnitOfWorkFactory                           |
| 22 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork.transaction               | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork             | Transaction<-TransactionalUnitOfWorkFactory                                  |
| 23 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | EventProcessorControlService<-AxonServerConfigurationEnhancer                |
| 24 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.event                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | AxonServerEventStorageEngineFactory<-AxonServerConfigurationEnhancer         |
| 25 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.command          | MetadataConverter<-CommandConverter                                          |
| 26 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                                | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.command          | AxonServerConfiguration<-AxonServerCommandBusConnector                       |
| 27 | axon-messaging-5.0.1        | org.axonframework.messaging.core.configuration                        | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.configuration | MessagingConfigurer<-EventProcessingConfigurer                               |
| 28 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling               | SequencingPolicy<-SimpleEventHandlingComponent                               |
| 29 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling               | SequentialPerAggregatePolicy<-SimpleEventHandlingComponent                   |
| 30 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling               | HierarchicalSequencingPolicy<-SimpleEventHandlingComponent                   |
| 31 | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling.sequencing                  | axon-messaging-5.0.1        | org.axonframework.messaging.eventhandling               | SequentialPolicy<-SimpleEventHandlingComponent                               |
| 32 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | AxonServerNonTransientRemoteQueryHandlingException<-ErrorCode                |
| 33 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | AxonServerRemoteQueryHandlingException<-ErrorCode                            |
| 34 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | AxonServerQueryDispatchException<-ErrorCode                                  |
| 35 | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector.query                          | axon-server-connector-5.0.1 | org.axonframework.axonserver.connector                  | AxonServerQueryBusConnector<-AxonServerConfigurationEnhancer                 |
| 36 | axon-modelling-5.0.1        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.1        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-EntityIdResolverDefinition                         |
| 37 | axon-modelling-5.0.1        | org.axonframework.modelling.entity.annotation                         | axon-modelling-5.0.1        | org.axonframework.modelling.annotation                  | AnnotatedEntityMetamodel<-AnnotationBasedEntityIdResolverDefinition          |
| 38 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork                           | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | ProcessingContext<-SubscribableEventSource                                   |
| 39 | axon-messaging-5.0.1        | org.axonframework.messaging.core.unitofwork                           | axon-messaging-5.0.1        | org.axonframework.messaging.core                        | ProcessingContext<-MessageDispatchInterceptorChain                           |



# Interface Segregation Candidates

Well known from [Design Principles and Design Patterns by Robert C. Martin](#), the *Interface Segregation Principle* suggests that software components should have narrow, focused interfaces rather than large, general-purpose ones. The goal is to minimize the dependencies between components and increase modularity, flexibility, and maintainability.

Smaller, focused and purpose-driven interfaces

- make it easier to modify individual components without affecting the rest of the system.
- make it clearer which client is affected by which change.
- don't force their clients to depend on methods they don't need.
- reduce the scope of changes since a change to one component doesn't affect others.
- lead to a more loosely coupled architecture that is easier to understand and maintain.

Reference: [Analyze java package metrics in a graph database](#)

## How to apply the results

If just one method of a type is used, especially in many places, then the result of this method can be used to call e.g. a method or construct an object instead of using the whole object and then just calling that single method.

If there are a couple of methods that are used for a distinct purpose, those could be factored out into a separate interface. The original type can extended/implement the new interface so that there are no breaking changes. Then all the callers, that use only this group of methods, can be changed to the new interface.

## Table 4 - Top 40 most used combinations of methods

The following table shows the top 40 most used combinations of methods of larger types that might benefit from applying the *Interface Segregation Principle*. The whole table can be found in the CSV report `Candidates_for_Interface_Segregation`.

|    | fullDependentTypeName                                                                                  | declaredMethods | calledMethodNames                                                                           | calledMethods | callerTypes |
|----|--------------------------------------------------------------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------|---------------|-------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 32              | [computeResourceIfAbsent]                                                                   | 1             | 7           |
| 1  | org.axonframework.messaging.core.unitofwork.UnitOfWork                                                 | 24              | [executeWithResult]                                                                         | 1             | 6           |
| 2  | org.axonframework.messaging.commandhandling.CommandBus                                                 | 6               | [dispatch]                                                                                  | 1             | 6           |
| 3  | org.axonframework.common.configuration.ComponentDefinition\$ComponentCreator                           | 17              | [createComponent]                                                                           | 1             | 5           |
| 4  | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 32              | [withResource]                                                                              | 1             | 4           |
| 5  | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [identifier, timestamp]                                                                     | 2             | 4           |
| 6  | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [timestamp]                                                                                 | 1             | 4           |
| 7  | org.axonframework.messaging.core.MessageStream\$Entry                                                  | 7               | [message]                                                                                   | 1             | 4           |
| 8  | org.axonframework.messaging.core.MessageStream\$Empty                                                  | 44              | [cast]                                                                                      | 1             | 3           |
| 9  | org.axonframework.messaging.core.DelayedMessageStream                                                  | 42              | [create]                                                                                    | 1             | 3           |
| 10 | org.axonframework.messaging.core.unitofwork.UnitOfWork                                                 | 24              | [execute]                                                                                   | 1             | 3           |
| 11 | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [identifier]                                                                                | 1             | 3           |
| 12 | org.axonframework.messaging.core.annotation.WrappedMessageHandlingMember                               | 13              | [handleSync]                                                                                | 1             | 3           |
| 13 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 32              | [putResource]                                                                               | 1             | 2           |
| 14 | org.axonframework.messaging.core.unitofwork.UnitOfWork                                                 | 25              | [executeWithResult]                                                                         | 1             | 2           |
| 15 | org.axonframework.common.configuration.AbstractComponent                                               | 24              | [describeTo]                                                                                | 1             | 2           |
| 16 | org.axonframework.test.fixture.AxonTestThenMessage                                                     | 22              | [exception, exceptionSatisfies]                                                             | 3             | 2           |
| 17 | org.axonframework.messaging.commandhandling.CommandMessage                                             | 19              | [andMetadata]                                                                               | 1             | 2           |
| 18 | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [withConvertedPayload]                                                                      | 1             | 2           |
| 19 | org.axonframework.messaging.eventhandling.EventMessage                                                 | 19              | [andMetadata]                                                                               | 1             | 2           |
| 20 | org.axonframework.messaging.eventstreaming.EventCriteria                                               | 12              | [tags]                                                                                      | 1             | 2           |
| 21 | org.axonframework.messaging.eventstreaming.OrEventCriteria                                             | 12              | [or]                                                                                        | 1             | 2           |
| 22 | org.axonframework.modelling.entity.ConcreteEntityMetamodel                                             | 11              | [forEntityClass]                                                                            | 1             | 2           |
| 23 | org.axonframework.modelling.entity.PolymorphicEntityMetamodel                                          | 11              | [forSuperType]                                                                              | 1             | 2           |
| 24 | org.axonframework.modelling.entity.annotation.AnnotatedEntityMetamodel                                 | 11              | [entityType]                                                                                | 1             | 2           |
| 25 | org.axonframework.eventsourcing.eventstore.EventStore                                                  | 9               | [transaction]                                                                               | 1             | 2           |
| 26 | org.axonframework.messaging.eventhandling.EventHandlingComponent                                       | 8               | [supportedEvents, supports, sequenceIdentifierFor]                                          | 3             | 2           |
| 27 | org.axonframework.messaging.eventhandling.processing.streaming.token.WrappedToken                      | 8               | [unwrapLowerBound]                                                                          | 1             | 2           |
| 28 | org.axonframework.messaging.core.MessageStream\$Entry                                                  | 7               | [map]                                                                                       | 1             | 2           |
| 29 | org.axonframework.messaging.core.unitofwork.UnitOfWork\$UnitOfWorkProcessingContext                    | 45              | [isCommitted, commit, isError, isCompleted, onError, whenComplete, isStarted, on]           | 8             | 1           |
| 30 | org.axonframework.messaging.core.EmptyMessageStream                                                    | 44              | [instance]                                                                                  | 1             | 1           |
| 31 | org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 43              | [isCompleted, hasNextAvailable]                                                             | 2             | 1           |
| 32 | org.axonframework.eventsourcing.eventstore.inmemory.InMemoryEventStorageEngine\$MapBackedMessageStream | 42              | [callback]                                                                                  | 1             | 1           |
| 33 | org.axonframework.messaging.core.DelayedMessageStream                                                  | 42              | [createSingle]                                                                              | 1             | 1           |
| 34 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [asCompletableFuture]                                                                       | 1             | 1           |
| 35 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [first]                                                                                     | 1             | 1           |
| 36 | org.axonframework.messaging.core.MessageStream\$Single                                                 | 42              | [cast]                                                                                      | 1             | 1           |
| 37 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 37              | [updateResource, removeResource, putResourceIfAbsent, putResource, computeResourceIfAbsent] | 6             | 1           |
| 38 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 36              | [updateResource, computeResourceIfAbsent]                                                   | 2             | 1           |
| 39 | org.axonframework.messaging.core.unitofwork.ProcessingContext                                          | 35              | [computeResourceIfAbsent]                                                                   | 1             | 1           |

## Package Usage

Table 5 - Types that are used by multiple packages

This table shows the top 40 packages that are used by the highest number of different packages. The whole table can be found in the CSV report `List_types_that_are_used_by_many_different_packages` .

|    | fullQualifiedDependentTypeName                                | dependentTypeName    | dependentTypeLabels                                                                                                                                                                                                                                                                                             | numberOfUsingPackages |
|----|---------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 0  | org.axonframework.messaging.core.unitofwork.ProcessingContext | ProcessingContext    | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...     | 57                    |
| 1  | org.axonframework.common.annotation.Internal                  | Internal             | [Type, File, Java, ByteCode, Annotation, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent...     | 47                    |
| 2  | org.axonframework.messaging.core.Message                      | Message              | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mar...     | 46                    |
| 3  | org.axonframework.common.infra.ComponentDescriptor            | ComponentDescriptor  | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...     | 39                    |
| 4  | org.axonframework.messaging.eventhandling.EventMessage        | EventMessage         | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedCompen...       | 35                    |
| 5  | org.axonframework.messaging.core.MessageStream                | MessageStream        | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWea...     | 35                    |
| 6  | org.axonframework.messaging.core.MessageType                  | MessageType          | [Type, File, Java, ByteCode, Record, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, M...     | 29                    |
| 7  | org.axonframework.messaging.core.QualifiedName                | QualifiedName        | [Type, File, Java, ByteCode, Record, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4T...     | 29                    |
| 8  | org.axonframework.common.configuration.Configuration          | Configuration        | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation25, Mark4TypeLouvainCommunity1...     | 28                    |
| 9  | org.axonframework.common.FutureUtils                          | FutureUtils          | [Type, File, Java, Class, ByteCode, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation27, Mark4TypeLouvainCommunity2, Mark...     | 26                    |
| 10 | org.axonframework.common.infra.DescribableComponent           | DescribableComponent | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0...     | 24                    |
| 11 | org.axonframework.messaging.core.MessageStream\$Empty         | MessageStream\$Empty | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity16, Mark4Typ...     | 24                    |
| 12 | org.axonframework.messaging.core.Context\$ResourceKey         | Context\$ResourceKey | [Type, File, Java, Class, ByteCode, GenericDeclaration, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyCon...     | 23                    |
| 13 | org.axonframework.common.BuilderUtils                         | BuilderUtils         | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation3, Mark4TypeLouvainCommunity1, Mark4TypeLeidenCommunity11, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut98, Mark4TypeHDBSCAN-1, Mark4TopAnomalyHub]                                                        | 21                    |
| 14 | org.axonframework.common.configuration.ComponentRegistry      | ComponentRegistry    | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation35, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity12, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut2, Mark4TypeHDBSCAN-1]                                        | 21                    |
| 15 | org.axonframework.common.Assert                               | Assert               | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityCostEffectivenessLazyForward, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation7, Mark4TypeLouvainCommunity5, Mark4Typ... | 20                    |
| 16 | org.axonframework.messaging.commandhandling.CommandMessage    | CommandMessage       | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity6, ...     | 20                    |
| 17 | org.axonframework.messaging.core.MessageStream\$Entry         | MessageStream\$Entry | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub,                                                                                                            | 19                    |

|    | fullQualifiedDependentTypeName                                                            | dependentTypeName                                  | dependentTypeLabels                                                                                                                                                                                                                                                                                          | numberOfUsingPackages |
|----|-------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                                                           |                                                    | Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity12, Mark4...                                                                                                                                                                                                       |                       |
| 18 | org.axonframework.messaging.core.annotation.ParameterResolverFactory                      | ParameterResolverFactory                           | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation67, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN56]                                                                     | 19                    |
| 19 | org.axonframework.messaging.core.MessageStream\$Single                                    | MessageStream\$Single                              | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mar...  | 18                    |
| 20 | org.axonframework.common.configuration.ConfigurationEnhancer                              | ConfigurationEnhancer                              | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation30, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity12, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN-1]                                                                    | 17                    |
| 21 | org.axonframework.messaging.core.Metadata                                                 | Metadata                                           | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut20, Ma...   | 16                    |
| 22 | org.axonframework.conversion.Converter                                                    | Converter                                          | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecompo...   | 15                    |
| 23 | org.axonframework.messaging.core.conversion.MessageConverter                              | MessageConverter                                   | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut18, Mark4TypeLocalClusteringCoefficient0.1225296442687747, Mark4TypeHDBSCAN153]              | 15                    |
| 24 | org.axonframework.messaging.eventhandling.processing.streaming.token.TrackingToken        | TrackingToken                                      | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityArticleRank, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation51, Mark4TypeLouvainCommunity2, Mark4TypeLeidenCommunity2, ... | 15                    |
| 25 | org.axonframework.common.AxonConfigurationException                                       | AxonConfigurationException                         | [Type, File, Java, Class, ByteCode, Throwable, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation10, Mark4TypeLouvainCommunity6, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN118]                                                              | 14                    |
| 26 | org.axonframework.messaging.core.MessageTypeResolver                                      | MessageTypeResolver                                | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut57, Mark4TypeHDBSCAN-1]                                                                      | 14                    |
| 27 | org.axonframework.common.ObjectUtils                                                      | ObjectUtils                                        | [Type, File, Java, Class, ByteCode, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut5, Mark4TypeHDBSCAN153]                                                                          | 14                    |
| 28 | org.axonframework.messaging.commandhandling.CommandBus                                    | CommandBus                                         | [Type, File, Java, ByteCode, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation8, Mark4TypeLouvainCommunity4, Mark4TypeLeidenCommunity6, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut0, Mark4TypeHDBSCAN106]                                                                      | 13                    |
| 29 | org.axonframework.messaging.queryhandling.QueryMessage                                    | QueryMessage                                       | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation44, Mark4TypeLouvainCommunity12, Mark4TypeLeidenCommunity4, Mark4TypeCoreDecomposition9, ...  | 13                    |
| 30 | org.axonframework.messaging.commandhandling.CommandResultMessage                          | CommandResultMessage                               | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityHyperlinkInducedTopicSearchAuthority, Mark4TopCentralityHyperlinkInducedTopicSearchHub, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation1, Mark4TypeLouvainCommunity0, Mark4TypeLeidenCommunity0, Mark4TypeCoreDecomposition9, M...   | 12                    |
| 31 | org.axonframework.common.configuration.ComponentDefinition                                | ComponentDefinition                                | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation38, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity12, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut42, Mark4TypeLocalClusteringCoefficient0.2028985507246377, Mark4...   | 12                    |
| 32 | org.axonframework.common.configuration.ComponentDefinition\$IncompleteComponentDefinition | ComponentDefinition\$IncompleteComponentDefinition | [Type, File, Java, ByteCode, GenericDeclaration, Interface, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation38, Mark4TypeLouvainCommunity11, Mark4TypeLeidenCommunity12, Mark4TypeCoreDecomposition9, Mark4TypeMaximumKCut75, Mark4TypeHDBSCAN123]                                              | 12                    |
| 33 | org.axonframework.messaging.core.Context                                                  | Context                                            | [Type, File, Java, ByteCode, Interface, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TopCentralityBetweenness, Mark4TopCentralityHarmonic, Mark4TopCentralityCloseness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation47, Mark4TypeLouvainCommunity12, Mark4TypeLeiden...  | 12                    |
| 34 | org.axonframework.common.ReflectionUtils                                                  | ReflectionUtils                                    | [Type, File, Java, Class, ByteCode, Mark4TopCentralityBetweenness, Mark4TypeWeaklyConnectedComponent0, Mark4TypeLabelPropagation20, Mark4TypeLouvainCommunity5, Mark4TypeLeidenCommunity7, Mark4TypeCoreDecomposition8, Mark4TypeMaximumKCut64, Mark4TypeHDBSCAN118, Mark4TopAnomalyBottleneck]              | 12                    |
| 35 | org.axonframework.common.AxonNonTransientException                                        | AxonNonTransientException                          | [Type, File, Java, Class, ByteCode, Mark4TopCentralityPageRank, Mark4TopCentralityArticleRank, Mark4TypeWeaklyConnectedComponent0,                                                                                                                                                                           | 11                    |

|    | fullQualifiedDependentTypeName                                | dependentTypeName         | dependentTypeLabels                                                                                                                                                                                                                                                                                                                 | numberOfUsingPackages |
|----|---------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|    |                                                               |                           | Mark4TypeLabelPropagation15,<br>Mark4TypeLouvainCommunity3,<br>Mark4TypeLeidenCommunity3,<br>Mark4TypeCoreDecomposition4,<br>Mark4TypeMaximumKCut33,<br>Mark4TypeLocalClusteringCoe...                                                                                                                                              |                       |
| 36 | org.axonframework.common.configuration.ComponentBuilder       | ComponentBuilder          | [Type, File, Java, ByteCode, GenericDeclaration, Interface,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation35,<br>Mark4TypeLouvainCommunity11,<br>Mark4TypeLeidenCommunity12,<br>Mark4TypeCoreDecomposition9,<br>Mark4TypeMaximumKCut86, Mark4TypeHDBSCAN112]                                                   | 11                    |
| 37 | org.axonframework.messaging.core.MessageHandlerInterceptor    | MessageHandlerInterceptor | [Type, File, Java, ByteCode, GenericDeclaration, Interface,<br>Mark4TopCentralityHyperlinkInducedTopicSearchAuthority,<br>Mark4TopCentralityHyperlinkInducedTopicSearchHub,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation86,<br>Mark4TypeLouvainCommunity11, Mark4...                                         | 11                    |
| 38 | org.axonframework.messaging.core.annotation.ParameterResolver | ParameterResolver         | [Type, File, Java, ByteCode, GenericDeclaration, Interface,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation14,<br>Mark4TypeLouvainCommunity6,<br>Mark4TypeLeidenCommunity7,<br>Mark4TypeCoreDecomposition7,<br>Mark4TypeMaximumKCut4, Mark4TypeHDBSCAN-1]                                                       | 11                    |
| 39 | org.axonframework.common.TypeReference                        | TypeReference             | [Type, File, Java, Class, ByteCode, GenericDeclaration,<br>Mark4TopCentralityPageRank,<br>Mark4TopCentralityArticleRank,<br>Mark4TopCentralityBetweenness,<br>Mark4TopCentralityHarmonic,<br>Mark4TopCentralityCloseness,<br>Mark4TypeWeaklyConnectedComponent0,<br>Mark4TypeLabelPropagation24,<br>Mark4TypeLouvainCommunity11,... | 11                    |

Table 6 - Packages that are used by multiple artifacts

This table shows the top 30 artifacts that only use a few (compared to all existing) packages of another artifact. The whole table can be found in the CSV report [ArtifactPackageUsage](#).



|    | artifactName                         | dependentArtifactName | packageName                                            | dependentPackage.fqn                        | dependentTypes | dependentPackageTypes | typeUsagePercentage |                  |
|----|--------------------------------------|-----------------------|--------------------------------------------------------|---------------------------------------------|----------------|-----------------------|---------------------|------------------|
| 0  | axon-modelling-5.0.1                 | axon-messaging-5.0.1  | org.axonframework.modelling.repository                 | org.axonframework.messaging.core            | 1              | 80                    | 0.012500            | [o               |
| 1  | axon-eventsourcing-5.0.1             | axon-messaging-5.0.1  | org.axonframework.eventsourcing.configuration          | org.axonframework.messaging.core            | 1              | 80                    | 0.012500            | [o               |
| 2  | axon-test-5.0.1                      | axon-messaging-5.0.1  | org.axonframework.test.matchers                        | org.axonframework.messaging.core            | 1              | 80                    | 0.012500            |                  |
| 3  | axon-eventsourcing-5.0.1             | axon-messaging-5.0.1  | org.axonframework.eventsourcing.configuration          | org.axonframework.messaging.core.annotation | 1              | 50                    | 0.020000            | [org.axonframewc |
| 4  | axon-spring-boot-autoconfigure-5.0.1 | axon-messaging-5.0.1  | org.axonframework.extension.springboot.autoconfigure   | org.axonframework.messaging.core.annotation | 1              | 50                    | 0.020000            | [org.axonframewo |
| 5  | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.core                       | org.axonframework.common.configuration      | 1              | 46                    | 0.021739            | [                |
| 6  | axon-eventsourcing-5.0.1             | axon-common-5.0.1     | org.axonframework.eventsourcing.annotation.reflection  | org.axonframework.common.configuration      | 1              | 46                    | 0.021739            | [                |
| 7  | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.core.unitofwork            | org.axonframework.common.configuration      | 1              | 46                    | 0.021739            | [org.axonframewo |
| 8  | axon-modelling-5.0.1                 | axon-common-5.0.1     | org.axonframework.modelling.entity.annotation          | org.axonframework.common.configuration      | 1              | 46                    | 0.021739            | [                |
| 9  | axon-eventsourcing-5.0.1             | axon-common-5.0.1     | org.axonframework.eventsourcing.annotation             | org.axonframework.common.configuration      | 1              | 46                    | 0.021739            | [                |
| 10 | axon-metrics-micrometer-5.0.1        | axon-messaging-5.0.1  | org.axonframework.extension.metrics.micrometer         | org.axonframework.messaging.core            | 2              | 80                    | 0.025000            |                  |
| 11 | axon-tracing-opentelemetry-5.0.1     | axon-messaging-5.0.1  | org.axonframework.extension.tracing.opentelemetry      | org.axonframework.messaging.core            | 2              | 80                    | 0.025000            |                  |
| 12 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.queryhandling.annotation   | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 13 | axon-test-5.0.1                      | axon-common-5.0.1     | org.axonframework.test.server                          | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 14 | axon-eventsourcing-5.0.1             | axon-common-5.0.1     | org.axonframework.eventsourcing.annotation             | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 15 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.core.timeout               | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 16 | axon-conversion-5.0.1                | axon-common-5.0.1     | org.axonframework.conversion.avro                      | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 17 | axon-spring-boot-autoconfigure-5.0.1 | axon-common-5.0.1     | org.axonframework.extension.springboot.autoconfigure   | org.axonframework.common                    | 1              | 34                    | 0.029412            | [c               |
| 18 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.eventhandling.processing   | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 19 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.eventhandling.sequencing   | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 20 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.queryhandling.tracing      | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 21 | axon-test-5.0.1                      | axon-common-5.0.1     | org.axonframework.test.fixture                         | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 22 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.eventhandling.tracing      | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 23 | axon-metrics-micrometer-5.0.1        | axon-common-5.0.1     | org.axonframework.extension.metrics.micrometer         | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 24 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.monitoring.configuration   | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 25 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.commandhandling.tracing    | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 26 | axon-tracing-opentelemetry-5.0.1     | axon-common-5.0.1     | org.axonframework.extension.tracing.opentelemetry      | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 27 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.monitoring                 | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 28 | axon-update-5.0.1                    | axon-common-5.0.1     | org.axonframework.update                               | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |
| 29 | axon-messaging-5.0.1                 | axon-common-5.0.1     | org.axonframework.messaging.core.unitofwork.annotation | org.axonframework.common                    | 1              | 34                    | 0.029412            |                  |

Table 8 - Duplicate package names across artifacts

This table shows the top 30 duplicate package names across artifacts. They are ordered by the number of duplicates descending.

This might lead to confusion, makes importing more error prone and might even lead to duplicate classes where only one of them will be loaded by the class loader. If a package is named the same way in two or more artifacts this even allows another artifact to access package protected classes, methods or members which might not be intended.

The whole table can be found in the CSV report `DuplicatePackageNamesAcrossArtifacts`.

| packageName | duplicates | artifactNames |
|-------------|------------|---------------|
|-------------|------------|---------------|

Table 9 - Annotated elements

This table shows 30 most used Java Annotations including some examples where they are used.

|    | annotationName                                                            | languageElement | numberOfAnnotatedElements |                                                                                                                                                                                                                            |
|----|---------------------------------------------------------------------------|-----------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0  | jakarta.annotation.NonNull                                                | Parameter       | 3382                      | [org.axonframework.extension.tracing.opentele<br>org.axonframework.extension.tracing.opentele<br>org.axonframework.extension.tracing.opentelemetry.MetadataContextGetter.get(1), org.axonframework.exte                    |
| 1  | jakarta.annotation.NonNull                                                | Method          | 833                       | [org.axonframework.common.util.ExecutorServiceFactory.createExecutorService, org.axonframework.comr<br>org.axonframework.common.ReflectionUtils.fieldsOf, org.axonframework.commo<br>org.axonframework.commo               |
| 2  | jakarta.annotation.Nullable                                               | Parameter       | 399                       | [org.axonframework.common.StringUtils.capitalize(0), org.axonframework.common<br>org.axonframework.common.ReflectionUtils.explicitlyUnequal(1), org.axonframework.common.lifecycle.LifecycleH<br>org.i                     |
| 3  | jakarta.annotation.Nullable                                               | Method          | 99                        | [org.axonframework.common.ReflectionUtils.declaringClass, org.axonframework<br>org.axonframework.common.configuration.AbstractComponent.name, org.axonframework.common.conf<br>org.axonframework.common.                   |
| 4  | org.axonframework.common.annotation.Internal                              | Class           | 75                        | [org.axonframework.common.configuration<br>org.axonframework.common.configuration.InstantiatedComponentDefinition, org.axonframework<br>org.axonframework.axonserver.connector.MetadataConverter, org.axc                  |
| 5  | jakarta.annotation.NonNull                                                | Field           | 70                        | [org.axonframework.commm<br><i>HandlerRegistration, handler, org. axon framework. common. configu</i><br>org.axonframework.axonserver.connector.command.AxonServerCommandBusCo<br>org.a)                                   |
| 6  | java.lang.FunctionalInterface                                             | Interface       | 55                        | [org.a)<br><i>SqlResultConverter, org. axon framework. common. jdbc. ConnectionProvider, org. axon frame</i><br>SqlFunction, org.axonframework.common.util.ExecutorServiceFactory, org.axonfr<br>org                       |
| 7  | java.lang.annotation.Retention                                            | Annotation      | 42                        | [org.axonframework.common.annotation.Internal, org.axonframework.common.Priority, org.axonframework<br>org.axonframework.modelling.annotation.InjectEntity, org.axonframework.mode<br>org.axonframe                        |
| 8  | java.lang.annotation.Target                                               | Annotation      | 42                        | [org.axonframework.common.annotation.Internal, org.axonframework.common.Priority, org.axonframework<br>org.axonframework.modelling.annotation.InjectEntity, org.axonframework.mode<br>org.axonframe                        |
| 9  | org.springframework.context.annotation.Bean                               | Method          | 36                        | [org.axonframework.extension.springboot.autoconfigure.AxonTimeoutAutoConfiguration.messag<br>org.axonframework.extension.springboot.autoconfigure.AxonTimeoutAutoConfiguration<br>org.axonframework.extension.             |
| 10 | org.axonframework.common.annotation.Internal                              | Constructor     | 19                        | [org.axonframework.eventsourcing.event<br>org.axonframework.messaging.eventhandling.processing.streaming.pooled.PooledStreaming<br>org.axonframework.messaging.eventhandling.processing.streaming.pooe                     |
| 11 | org.springframework.boot.autoconfigure.condition.ConditionalOnMissingBean | Method          | 18                        | [org.axonframework.extension.springboot.autoconfigure.JpaAutoCo<br>org.axonframework.extension.springboot.autocor<br>org.axonframework.extension.springboot.autoconfigure.JpaAutoConfiguration.persistenceExci             |
| 12 | org.springframework.boot.autoconfigure.AutoConfiguration                  | Class           | 18                        | [org.axonframework.extension.springboot.autocci<br>org.axonframework.extension.springboot.autoconfigure.Correl<br>org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguration, org.axi           |
| 13 | java.lang.annotation.Documented                                           | Annotation      | 15                        | [org.axonframework.common.annotation.Internal, org.axonframework.mode<br>org.axonframework.eventsourcing.annotation.EventSourcingHandler, org.axonframework.extension.springt<br>org.axonframeworkworl                     |
| 14 | org.axonframework.common.annotation.Internal                              | Interface       | 14                        | [org.axonframework.common.jdbc.ConnectionProvider, org.axonframework<br>org.axonframework.modelling.entity.annotation.AnnotatedEntityMetamodelFactory, org.axonframework.eventsx<br>org.axonframework.ev                   |
| 15 | com.fasterxml.jackson.annotation.JsonProperty                             | Parameter       | 12                        | [org.axonframework.messaging.eventhandling.processing.streaming.token<br>org.axonframework.messaging.eventhandling.processing.streaming.token<br>org.axonframework.messaging.eventhandling.processing.strei                |
| 16 | org.axonframework.common.Priority                                         | Class           | 12                        | [org.axonframework.test.Fixtur<br>org.axonframework.messaging.eventhandling.annotation.TimestampParameterResolverFactory, org.axonframeworkworl                                                                            |
| 17 | org.axonframework.common.annotation.Internal                              | Method          | 10                        | [org.axonframework.messaging.eventhandling.conversion<br>org.axonframework.messaging.eventhandling.processing.streaming.token<br>org.axonframework.messaging.eventhandling.processing.streaming.token                      |
| 18 | org.springframework.boot.context.properties.ConfigurationProperties       | Class           | 10                        | [org.axonframework.axonserver.connector.AxonServerConfiguration, org.axonframework.extension.springboot<br>org.axonframework.extension.springboot.TokenStoreProperties, org.axonframework.extension.spr                    |
| 19 | org.springframework.boot.context.properties.EnableConfigurationProperties | Class           | 9                         | [org.axonframework.extension.springboot.autocci<br>org.axonframework.extension.springboot.springb<br>org.axonframework.extension.springboot.autoconfigure<br>org.axonframework.extension.                                  |
| 20 | org.springframework.boot.autoconfigure.condition.ConditionalOnProperty    | Class           | 8                         | [org.axonframework.extension.springboot.autocci<br>org.axonframework.extension.springboot.autoconfigure.Avro<br>org.axonframework.extension.springboot.autoconfigure.Avro                                                  |
| 21 | jakarta.persistence.Basic                                                 | Field           | 8                         | [org.axonframework.eventsourcing.event<br>org.axonframework.eventsourcing.eventst<br>org.axonframework.eventsourcing.eventstore<br>org.axonframework.eventsourcing.evi                                                     |
| 22 | org.springframework.boot.context.properties.NestedConfigurationProperty   | Field           | 8                         | [org.axonframework.ext<br><i>MessageHandlerTimeoutProperties, queries, org. axon framework. extension. .</i><br>MessageHandlerTimeoutProperties.deadlines, org.axonframework.extension.springboot.TimeoutProperties\$Messe |
| 23 | java.lang.SafeVarargs                                                     | Constructor     | 7                         | [org.axonframework.test.matchers.SequenceMatcher.<init>-, org.axonframework<br>org.axonframework.test.matchers.ExactSequenceMatcher.<init>-, org.axonframework.test.n<br>org.axonframework.test.matchers                   |
| 24 | org.springframework.boot.autoconfigure.condition.ConditionalOnClass       | Class           | 7                         | [org.axonframework.extension.springboot.autoconfigure<br>org.axonframework.extension.springboot<br>org.axonframework.extension.springboot.autoconfigure.AxonServerAutoConfiguration, org.axonframework.extension           |
| 25 | org.springframework.context.annotation.Conditional                        | Method          | 6                         | [org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoCo<br>org.axonframework.extension.springboot.autoconfigure.AvroSchemaStoreAutoConfiguratio<br>org.axonframework.extension.springboot.au           |
| 26 | org.springframework.boot.context.properties.bind.DefaultValue             | Parameter       | 6                         | [org.axonframework.extension.springboot.JpaEventStorageEng<br>org.axonframework.extension.springboot.JpaEventStorageEng<br>org.axonframework.extension.springboot.JpaEventStorageEngineConfigura                           |
| 27 | com.fasterxml.jackson.annotation.JsonCreator                              | Constructor     | 6                         | [org.axonframework.messaging.eventhandling.processing.streaming.ti<br>org.axonframework.messaging.eventhandling.processing.streaming.strai<br>org.axonframework.messaging.eventhandling.processing.streaming.token         |
| 28 | java.beans.ConstructorProperties                                          | Constructor     | 6                         | [org.axonframework.messaging.eventhandling.processing.streaming.ti<br>org.axonframework.messaging.eventhandling.processing.streaming.ti<br>org.axonframework.messaging.eventhandling.processing.streaming.strai            |
| 29 | org.springframework.boot.autoconfigure.AutoConfigureBefore                | Class           | 6                         | [org.axonframework.extension.springboot.autoconfigure.Correl<br>org.axonframework.extension.springboot.autoconfigure<br>org.axonframework.extension.springboot.autoconfigure.ConvertAutoConfiguration, org.axoni           |

Table 10 - Distance distribution between dependent files

This table shows the file directory distance distribution between dependent files. Intuitively, the distance is given by the fewest number of change directory commands needed to navigate between a file and a dependency it uses. Those are aggregate to see how many dependent files are in the same directory, how many are just one change directory command apart, and so on.



| dependency.fileDistanceAsFewestChangeDirectoryCommands | numberOfDependencies | numberOfDependencyUsers | numberOfDependencyProviders |     |
|--------------------------------------------------------|----------------------|-------------------------|-----------------------------|-----|
| 0                                                      | 0                    | 2080                    | 854                         | 896 |
| 1                                                      | 1                    | 96                      | 82                          | 41  |
| 2                                                      | 2                    | 2067                    | 606                         | 417 |
| 3                                                      | 4                    | 2019                    | 650                         | 309 |